

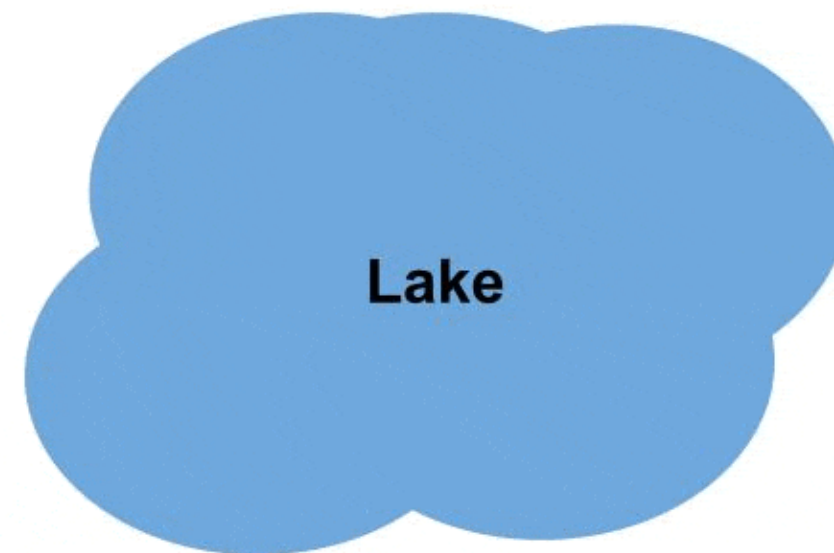
Ingesting Data

INTRODUCTION TO DATABRICKS SQL

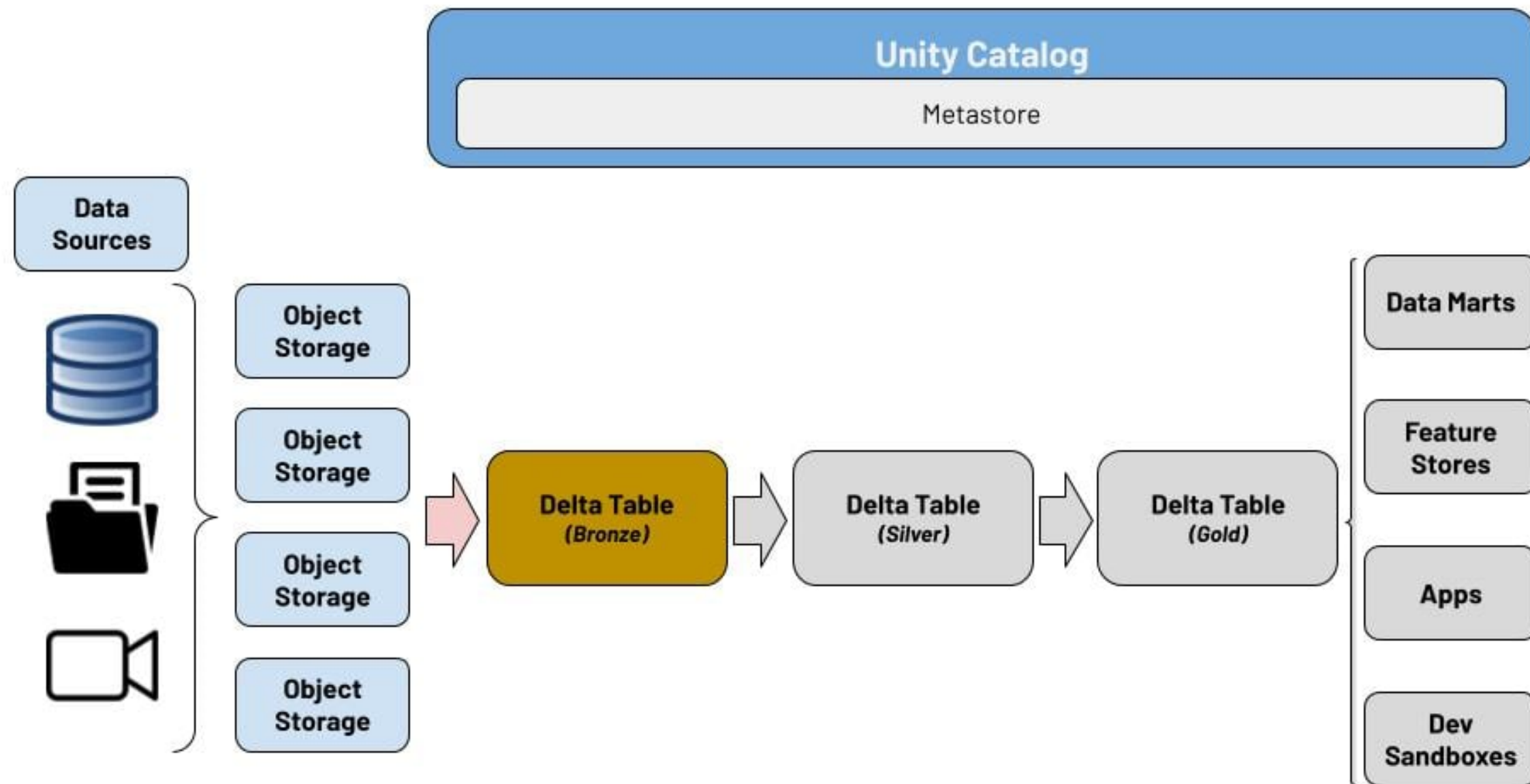


Kevin Barlow
Data Manager

Motivation



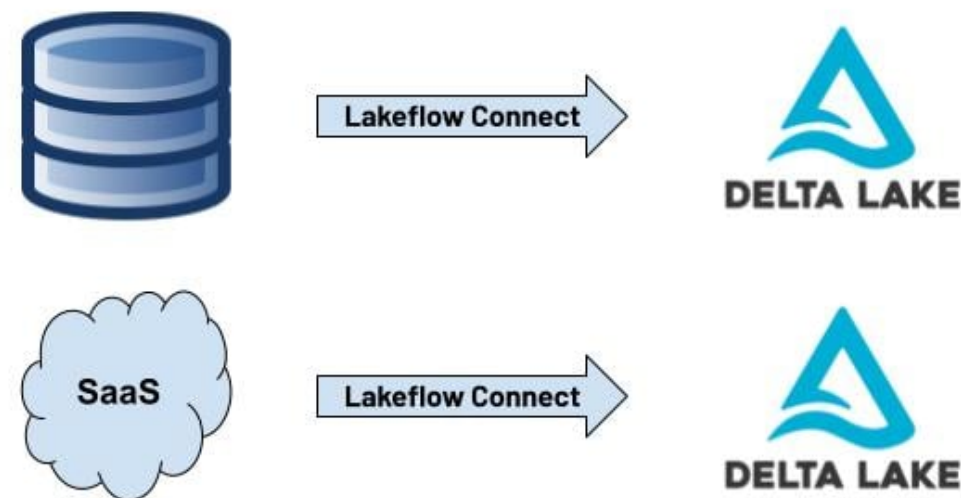
Creating the lakehouse



GUI-based options

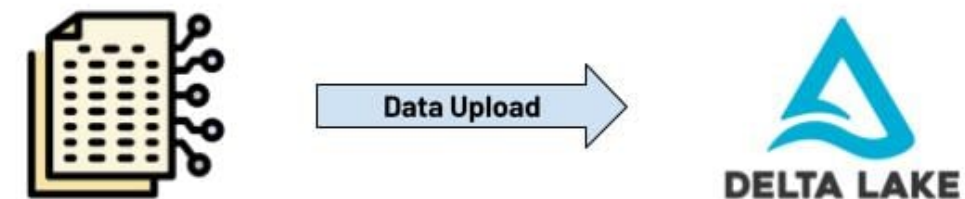
Lakeflow Connect

- Built-in connectors for ingesting data
 - Databases
 - SaaS Applications
- Creates pipelines to keep data up to date



Data upload

- Manually upload your files
 - CSV, Parquet, etc.
- Quickly create new Delta tables
- Great for ad hoc data upload



Bringing data into the lakehouse

COPY INTO

- Copy data from cloud object storage directly into Delta tables
- Better for more static datasets
- Can run natively in *SQL Editor*

```
COPY INTO my_table
FROM '/path/to/files'
FILEFORMAT = PARQUET
FORMAT_OPTIONS ('mergeSchema' = 'true')
COPY_OPTIONS ('mergeSchema' = 'true')
```

Auto Loader

- Automatically ingests new data files from cloud storage
- Better for larger and changing datasets
- Leverages *Delta Live Tables* in SQL

```
CREATE TABLE customers
AS SELECT *
FROM cloud_files(
  "/path/to/files",
  "csv")
```

Let's practice!

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Hydrating the lakehouse

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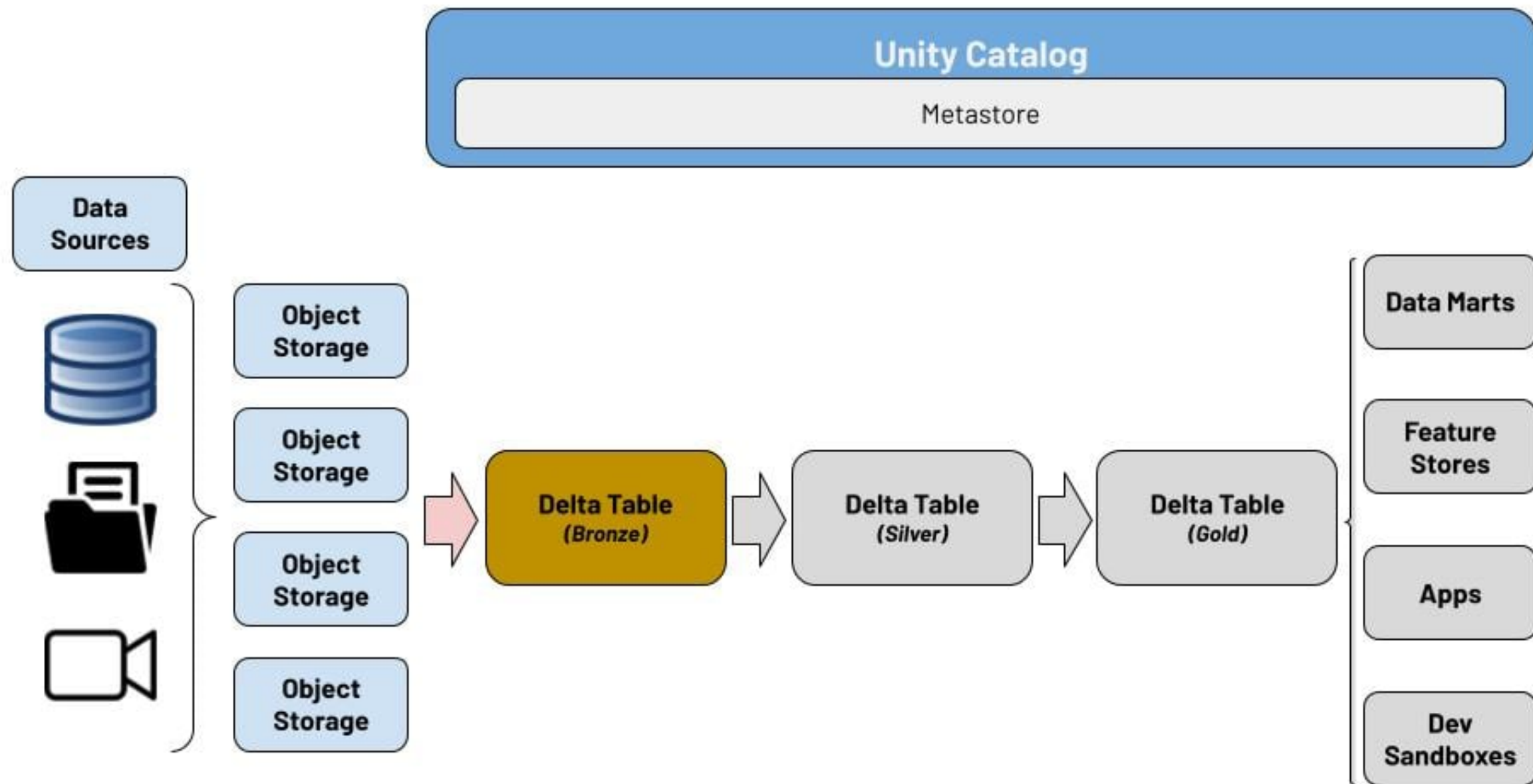
Transforming data

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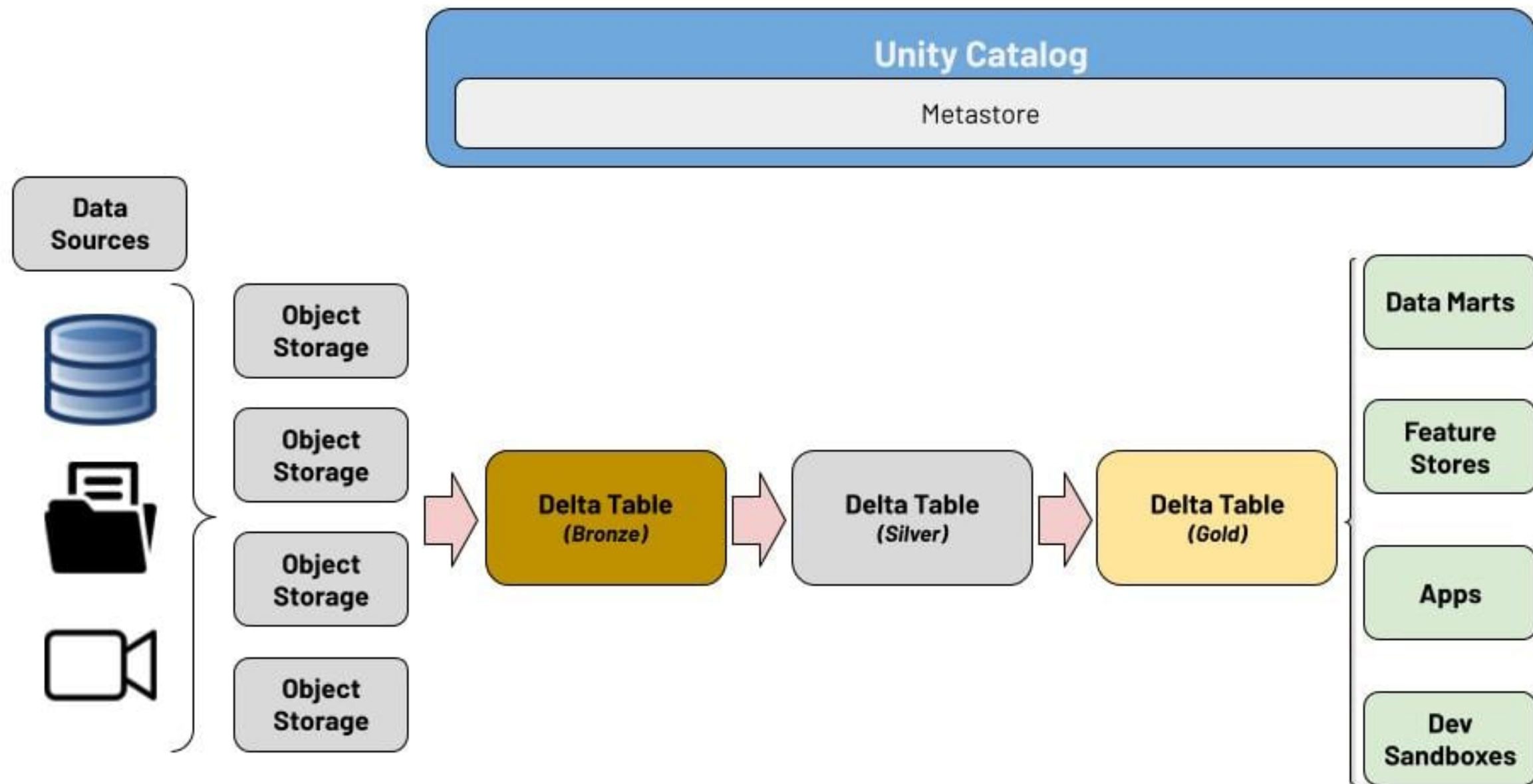


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Motivation



Transformation in the lakehouse



Cleaning and transforming data

- Cleaning data in the Bronze (raw) layer into the Silver (analytics-ready) layer
- Important step for downstream data tables
- Common activities
 - Removing NULL values
 - Standardize values
 - Adjusting data types

```
CREATE TABLE silver_layer AS (  
  SELECT DISTINCT c.id,  
                  c.last_name || ', ' || c.first_name  
                  AS name,  
                  format(s.date, 'YYYY-mm-dd')  
                  AS sale_date,  
                  round(s.price, 2)  
                  AS sale_price  
                  s.item_name  
  FROM sales_data s  
  LEFT JOIN contacts c on c.id = s.id)
```

Aggregating data

- Combining and simplifying data from Silver layer into Gold (BI-ready) layer
- Meant for a specific business intelligence need
 - Great candidate for views
- Common activities
 - Removing extraneous columns
 - Aggregating across dimensions
 - Calculating metrics / KPIs

```
CREATE VIEW q3_revenue AS (  
  SELECT sum(revenue) AS total_rev,  
         count(*) AS total_count,  
         total_rev / total_count AS avg_sale,  
         category,  
         item  
  FROM silver_layer  
  WHERE date BETWEEN '2024-07-01'  
         AND '2024-09-30'  
  GROUP BY category, item)
```

Automating tasks

The screenshot displays the Databricks web interface. The top navigation bar includes the Databricks logo, a search bar, and user information. The left sidebar contains a menu with options like New, Workspace, Repos, Recents, Data, Workflows, Compute, SQL, SQL Editor, Queries, Dashboards, Alerts, Query History, SQL Warehouses, Data Engineering, Delta Live Tables, Machine Learning, and Experiments. The main content area shows a workflow titled 'Twitter-DLT-ML-Workflow' under the 'Jobs' tab. The workflow is visualized as a sequence of three tasks: 'Ingest_Tweets' (using a Delta Live Table Pipeline), 'Process_Tweets' (using a Delta Live Table Pipeline), and 'Analyze_Tweets' (using a job ML-Huggingface_cluster). A large blue plus sign is centered below the workflow graph. A message at the bottom states 'No task selected' and 'Choose a task from the graph to edit its properties'.

Workflow: Twitter-DLT-ML-Workflow

Tasks:

- Ingest_Tweets**
 - ...-dlit-hugface-v1/Twitter-Stream-S3
 - job ML-Huggingface_cluster
- Process_Tweets**
 - Delta Live Table Pipeline
- Analyze_Tweets**
 - ...face-v1/Twitter-SentimentAnalysis
 - job ML-Huggingface_cluster

Message: No task selected. Choose a task from the graph to edit its properties.

Let's practice!

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Creating a coffee data layer

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Let's practice!

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